

ITA0515
Computer Vision
Lab Programs
(05 - 10)

Y. Jayesh Ranjan

192472201

EXP6

Aim

To read a video file in Python and display it in slow motion and fast motion using OpenCV.

Algorithm

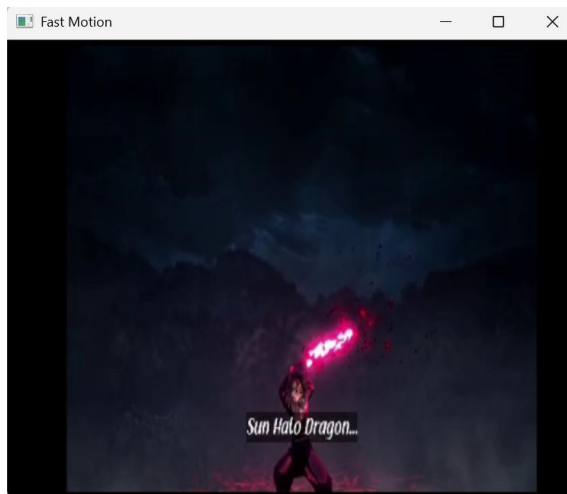
1. Import the OpenCV library.
2. Load the video file.
3. Read each frame from the video.
4. Display the frame in slow motion.
5. Display the frame in fast motion.
6. Press **Q** to stop the video.
7. Release the video and close all windows.

Code

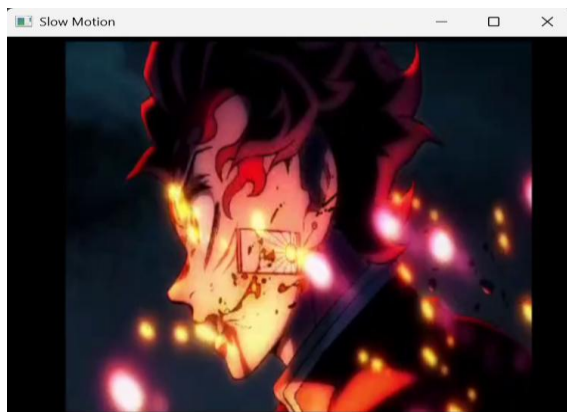
```
import cv2
cap = cv2.VideoCapture("sample.mp4")
while True:
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow("Video", frame)
    cv2.waitKey(80)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the program to display a video in slow motion and fast motion was executed successfully.

EXP7

Aim

To capture video from a webcam and display it using OpenCV.

Algorithm

1. Import the OpenCV library.
 2. Open the webcam.
 3. Capture video frame by frame.
 4. Display the video.
 5. Press **Q** to stop.
 6. Release the webcam and close all windows.
-

Code

```
import cv2

cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    cv2.imshow("Webcam", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
```

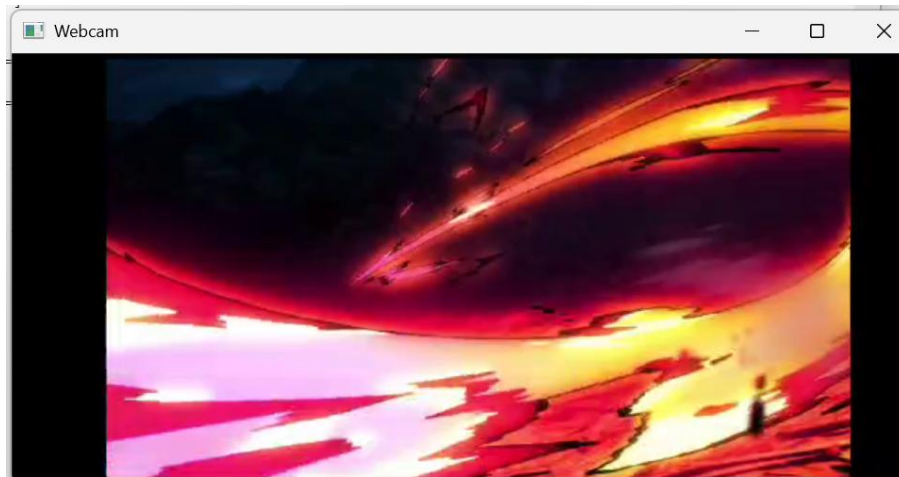
```
cap.release()
```

```
cv2.destroyAllWindows()
```

Input

webcam

Output



Result

Thus, the program to capture and display webcam video was executed successfully.

EXP8

Aim

To resize an image into bigger and smaller sizes using OpenCV.

Algorithm

1. Import the OpenCV library.
2. Read the input image.
3. Find the image dimensions.
4. Resize the image to a bigger size.
5. Resize the image to a smaller size.
6. Display both resized images.
7. Save the output images.

Code

```
import cv2

image = cv2.imread("image1.jpg")
height, width = image.shape[:2]

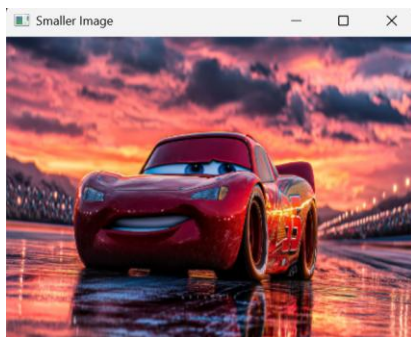
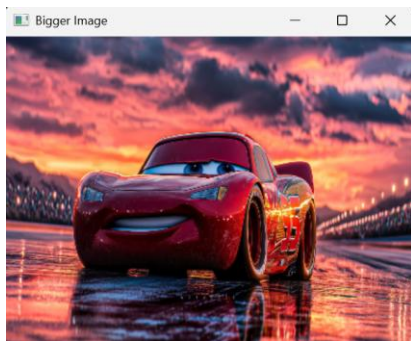
bigger = cv2.resize(image, (width * 2, height * 2))
smaller = cv2.resize(image, (width // 2, height // 2))

cv2.imshow("Original Image", image)
cv2.imshow("Bigger Image", bigger)
cv2.imshow("Smaller Image", smaller)
cv2.imwrite("bigger_image.jpg", bigger)
cv2.imwrite("smaller_image.jpg", smaller)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the program to resize an image into bigger and smaller sizes was executed successfully.

EXP9

Aim

To rotate an image in clockwise and counter-clockwise directions using OpenCV.

Algorithm

1. Import the OpenCV library.
 2. Read the input image.
 3. Rotate the image clockwise.
 4. Rotate the image counter-clockwise.
 5. Display the original and rotated images.
 6. Save the rotated images.
-

Code

```
import cv2

image = cv2.imread("image1.jpg")

clockwise = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)

counter_clockwise = cv2.rotate(image, cv2.ROTATE_90_COUNTERCLOCKWISE)

display_original = cv2.resize(image, (400, 300))

display_clockwise = cv2.resize(clockwise, (400, 300))

display_counter = cv2.resize(counter_clockwise, (400, 300))

cv2.imshow("Original Image", display_original)

cv2.imshow("Clockwise Rotation", display_clockwise)

cv2.imshow("Counter Clockwise Rotation", display_counter)

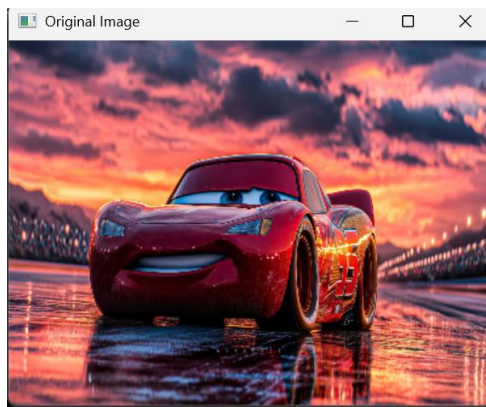
cv2.imwrite("clockwise.jpg", clockwise)

cv2.imwrite("counter_clockwise.jpg", counter_clockwise)

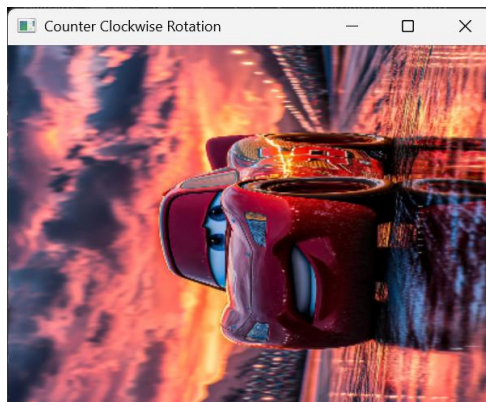
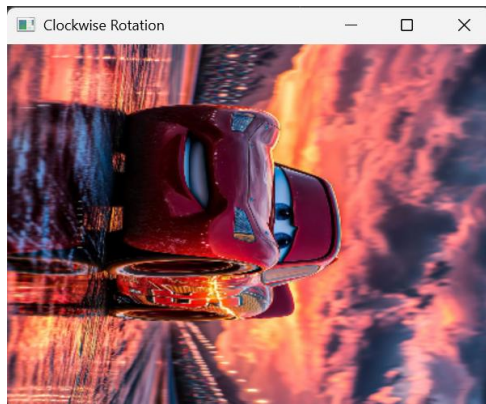
cv2.waitKey(0)

cv2.destroyAllWindows()
```


Input



Output



Result

Thus, the program to rotate an image in clockwise and counter-clockwise directions was executed successfully.

EXP10

Aim

To move an image from one position to another on the screen using OpenCV.

Algorithm

1. Import the OpenCV library.
2. Read the input image.
3. Resize the image for display.
4. Display the image.
5. Move the image window to a new position.
6. Wait for a key press and close the window.

Code

```
import cv2  
  
image = cv2.imread("image1.jpg")  
image = cv2.resize(image, (400, 300))  
cv2.imshow("Original Image", image)  
cv2.moveWindow("Original Image", 300, 200)  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

Output



Result

Thus, the program to move an image from one position to another was executed successfully.